

DEPARTMENT OF MATHEMATICS, IIT KHARAGPUR
Mid Spring Semester Examination 2023-2024

Subject Name: **Transform Calculus**, Subject Code: **MA20202**, FM = 30M, Duration 2 Hrs
Dept: AE, CH, CS, CY, EC, EE, EX, IE, IM, MA, MT, ME, MF

Specific charts, graph paper, log book etc., required ----- No

Special instruction: Answer booklet will be provided which is meant for this subject only.

- ❖ Please write your section in the answer booklet in the space provided. Otherwise your copy may be misplaced.
- ❖ Attempt all four questions. Symbols have their usual meaning if not explained.
- ❖ Answer each question in the space provided for that question only in the answer booklet provided.
- ❖ Do the rough work in the space provided in the answer booklet only.

1. [2+2+3+1=8M]

a) Verify whether the function $f(t) = e^{t^2} \log_e |t| + t^n, n \in \mathbb{Z}^+$ is of exponential order or not.

b) With the help of Laplace transform, obtain the value of the integral

$$\int_0^{\infty} t^3 e^{-t} \sin(t) dt$$

c) Obtain the Laplace Transform of the function $f: [0, \infty) \rightarrow \mathbb{R}$ which is defined by

$$f(t) = \begin{cases} t+1 & \text{for } 0 \leq t \leq 2, \\ 3 & \text{for } t > 2. \end{cases}$$

And, hence deduce the value of $L\{f'(t)\}$ at $s = 1$.

d) Suppose Laplace transform of a function $f(t)$ is given by

$$F(s) = \frac{18}{s(s^2 + 36)}$$

Without finding $f(t)$ by inverting $F(s)$, find $f(0)$. Write clearly the statement of the property/theorem that you are using here.

2. [3+2+2=7M]

a) Evaluate $L\{t \operatorname{erf}(2\sqrt{t})\}$, where the error function is defined as $\operatorname{erf}(t) = \frac{2}{\sqrt{\pi}} \int_0^t e^{-x^2} dx$.

b) Find $L^{-1}\left\{\frac{e^{-1/s}}{s}\right\}$ and write your answer in terms of Bessel function of zero order.

c) Find $L^{-1}\left\{\ln\left(1 + \frac{1}{s^2}\right)\right\}$.

$$\sqrt{s^3(s+1)}$$

3. [3+3+2=8M]

a) Use Laplace transform to find the solution of the initial value problem
 $x''(t) + tx'(t) - 2x = 2, x(0) = x'(0) = 0$

b) Find the solution of the integral equation
 $g(t) = \int_0^t f'(\tau)(t - \tau)^{-\alpha} d\tau, 0 < \alpha < 1$

c) Using Laplace transform with respect to t , evaluate the improper integral

$$f(t) = \int_0^\infty \frac{\sin(tx)}{x(1+x^2)} dx$$

4. [3+2+2=7M]

a) Find two solutions of the integral equation

$$y(t) = \frac{1}{2} \sin 2t + \int_0^t y(u)y(t-u) du$$

Write the solutions in terms of Bessel function/Dirac Delta function.

b) Given $f(t) = n, n \leq t < n+1, n = 0, 1, 2, 3, \dots$. Find $L\{f(t)\}$.

c) Solve the differential equation

$$y''(t) + 4y(t) = \delta(t-2), y(0) = 0, y'(0) = 1$$

where δ is the Dirac delta function. Use Heaviside step function to write the answer.

*****The End*****

$$J_0(x) = \sqrt{\frac{2}{\pi x}} \cos\left(\sqrt{x^2 - a^2}\right)$$